

### **ER diagram to Relational schema**

```
CREATE TABLE TimePeriod (  
    FL_DATE DATE PRIMARY KEY,  
    YEAR SMALLINT NOT NULL,  
    MONTH VARCHAR(20) NOT NULL,  
    DAY_OF_MONTH TINYINT NOT NULL,  
    DAY_OF_WEEK VARCHAR(20) NOT NULL,  
    QUARTER VARCHAR(20) NOT NULL  
);
```

```
CREATE TABLE Airlines(  
    OP_CARRIER_AIRLINE_ID INT PRIMARY KEY,  
    OP_UNIQUE_CARRIER VARCHAR(100) NOT NULL  
);
```

```
CREATE TABLE Airports(  
    AIRPORT_ID INT PRIMARY KEY,  
    IATA_CODE VARCHAR(20) NOT NULL,  
    CITY_NAME VARCHAR(80) NOT NULL,  
    STATE_ABR VARCHAR(20) NOT NULL,  
    FOREIGN KEY (STATE_ABR) REFERENCES StateNames(STATE_ABR)  
);
```

```
CREATE TABLE Flights (  
    FLIGHT_ID INT PRIMARY KEY,  
    OP_CARRIER_FL_NUM INT NOT NULL,  
    TAIL_NUM VARCHAR(20) NULL,  
    OP_CARRIER_AIRLINE_ID INT NOT NULL,
```

ROUTE\_ID INT NOT NULL,  
FL\_DATE DATE NOT NULL,  
CANCELLED TINYINT NOT NULL,  
DIVERTED TINYINT NOT NULL,  
FOREIGN KEY (OP\_CARRIER\_AIRLINE\_ID) REFERENCES  
Airlines(OP\_CARRIER\_AIRLINE\_ID),  
FOREIGN KEY (ROUTE\_ID) REFERENCES Routes(ROUTE\_ID),  
FOREIGN KEY (FL\_DATE) REFERENCES TimePeriod(FL\_DATE)  
);

CREATE TABLE FlightPerformance (  
FLIGHT\_ID INT PRIMARY KEY,  
CRS\_DEP\_TIME TIME NOT NULL,  
DEP\_TIME TIME,  
DEP\_DELAY INT,  
DEP\_DELAY\_NEW INT,  
DEP\_DEL15 TINYINT,  
TAXI\_OUT SMALLINT,  
WHEELS\_OFF TIME,  
WHEELS\_ON TIME,  
TAXI\_IN SMALLINT,  
CRS\_ARR\_TIME TIME,  
ARR\_TIME TIME,  
ARR\_DELAY INT,  
ARR\_DELAY\_NEW INT,

ARR\_DEL15 TINYINT,  
CRS\_ELAPSED\_TIME SMALLINT,  
ACTUAL\_ELAPSED\_TIME SMALLINT,  
AIR\_TIME SMALLINT,  
CANCELLATION\_CODE CHAR(50),  
DIV\_AIRPORT\_LANDINGS TINYINT,  
DIV\_REACHED\_DEST TINYINT,  
DIV\_ACTUAL\_ELAPSED\_TIME TINYINT,  
DIV\_ARR\_DELAY INT,  
DIV\_DISTANCE INT,  
FOREIGN KEY (FLIGHT\_ID) REFERENCES Flights(FLIGHT\_ID),  
FOREIGN KEY (CANCELLATION\_CODE) REFERENCES  
CancellationReason(CANCELLATION\_CODE)  
);

CREATE TABLE Routes (  
ROUTE\_ID INT PRIMARY KEY AUTO\_INCREMENT,  
ORIGIN\_AIRPORT\_ID INT NOT NULL,  
DEST\_AIRPORT\_ID INT NOT NULL,  
DISTANCE INT NOT NULL,  
FOREIGN KEY (ORIGIN\_AIRPORT\_ID) REFERENCES Airports(AIRPORT\_ID),  
FOREIGN KEY (DEST\_AIRPORT\_ID) REFERENCES Airports(AIRPORT\_ID),  
);

CREATE TABLE FlightDelay (  
DELAY\_ID TINYINT NOT NULL,

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FLIGHT_ID INT NOT NULL,  
  
MINUTES INT,  
  
PRIMARY KEY (DELAY_ID, FLIGHT_ID),  
  
FOREIGN KEY (DELAY_ID) REFERENCES DelayType(DELAY_ID),  
  
FOREIGN KEY (FLIGHT_ID) REFERENCES Flights(FLIGHT_ID)  
  
);  
  
CREATE TABLE FlightDiversion (  
  
FLIGHT_ID INT NOT NULL,  
  
DIVERSION_NUMBER TINYINT NOT NULL,  
  
DIV_AIRPORT_ID INT NOT NULL,  
  
DIV_TOTAL_GTIME SMALLINT NULL,  
  
PRIMARY KEY (FLIGHT_ID, DIVERSION_NUMBER),  
  
FOREIGN KEY (FLIGHT_ID) REFERENCES Flights(FLIGHT_ID),  
  
FOREIGN KEY (DIV_AIRPORT_ID) REFERENCES Airports(AIRPORT_ID)  
  
);
```

```
CREATE TABLE StateNames (  
  
STATE_ABR CHAR(2) PRIMARY KEY,  
  
STATE_NM VARCHAR(50) NOT NULL  
  
);
```

```
CREATE TABLE AirlineNames (  
  
OP_CARRIER_AIRLINE_ID INT PRIMARY KEY,  
  
NAME VARCHAR(150) NOT NULL  
  
);
```

```
CREATE TABLE AirportNames (  
  

```

```
AIRPORT_ID INT PRIMARY KEY,  
NAME VARCHAR(100) NOT NULL  
);
```

```
CREATE TABLE DelayType (  
DELAY_ID TINYINT PRIMARY KEY,  
NAME VARCHAR(50) NOT NULL  
);
```

```
CREATE TABLE CancellationReason (  
CANCELLATION_CODE CHAR(1) PRIMARY KEY,  
DESCRIPTION VARCHAR(50) NOT NULL  
);
```